



200 series

SH-UVCOD200

COD digital probe

User manual

Reading instructions

Information for users

Thank you very much for choosing to use our SH-UVCOD200 COD online digital probe (hereinafter referred to as: COD200 probe). Before using this product, please read this user manual carefully. This manual covers important information and data for the use of the product, which must be strictly observed by the user to ensure the normal operation of the COD200 probe. The information will help you to use the product correctly and to obtain accurate analysis results.

General description

This manual provides a detailed description of the installation, commissioning, operation and maintenance of the COD200 probe, as well as describing the measurement principle, instrument composition and performance characteristics of the COD200 probe.

This manual instructs the user on the proper installation and operation of the COD200 probe, as well as preventive maintenance work on the COD200 probe to guarantee the continuous and reliable operation of this sensor.

The products described in this manual have been rigorously inspected before leaving the factory to ensure top quality. In order to ensure safe, high-quality operation and to obtain correct analysis results, the user must strictly follow the instructions described in this manual.

This manual provides all the information required for the proper use of the COD200 probe. It provides an accurate reference for use by technicians with specialized training or knowledge of the instrument's operational controls. Before operating the instrument, make sure that the safety and warning messages mentioned in this manual are correctly understood and applied to the actual operation.

Delivery and transport

Shipment requirements are in accordance with the terms of the ordering contract.

Please read the appropriate information on the packaging material carefully when opening the box to ensure that the opened goods are complete and undamaged. Please try to retain the outer packaging of the product in case you need to return the meter or parts.

Warranty and maintenance

1) SYSTEAL warrants all Products manufactured to be free from defects in material and workmanship under normal use and service. SYSTEAL's obligation under any warranty is limited to and Buyer's exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEAL's factory, of any defective parts, when returned to SYSTEAL by Buyer, transportation prepaid, which SYSTEAL's examination discloses to have been factory defective. The time limit of this warranty is 12 (twelve) months from date of shipment of new Products unless otherwise specified by SYSTEAL. SYSTEAL shall not be liable for consequential damages, in any event. SYSTEAL will decide at its only discretion whether to repair or replace a defective part.

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3) THESE WARRANTIES ARE IN LIEU OF ANY WARRANTY OF MERCHANTABILITY, OR FITNESS, OR ANY OTHER WARRANTY, EXPRESSED OR IMPLIED.

4) Expendable Items: this Warranty does not cover expendable or consumable items such as lamps, fuses, parts in contact with fluids, etc.

5) SYSTEALOG has reasonable period of time for replacements/repairs, and cannot liable for any compensation may arise in relation to the Product

6) SYSTEALOG only obligation in case of defects, lack of quality or non-conformity of the Products will be repairing or replacing the defective parts. It is agreed that the above-mentioned warranty (i.e.: the obligation to repair or replace the defective parts) is in lieu of any other legal guarantee or liability with the exclusion of any other SYSTEALOG liability (whether contractual or non-contractual) which may anyhow arise out of or in relation with the Products supplied (e.g. compensation of damages, loss of profit, recall campaigns, etc.).

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- b) Damage caused by force majeure (earthquakes, lightning strikes, floods, etc.);
- c) Unauthorized alterations within the product without permission;
- d) Failure to use the product in accordance with the user's manual and training provisions, causing damage to the product.
- e) On products not manufactured by SYSTEALOG but sold by SYSTEALOG as results of a distribution contract or as occasional resale, apply the warranty conditions of the manufacturer of the product.
- f) Any warranty request, within specified limits, must be notified, the SYSTEALOG using the form "Warranty Claim" by fax, mail or registered mail, within seven (7) days from the event of the defect. Failing such notification, the Purchaser's right to claim the defects will be forfeited.

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1 Instrument overview

1.1 Instrument Introduction

The COD digital probe is able to monitor three parameters: temperature, turbidity and COD value in water in real time, and the appearance of the device is shown in Fig. 1.1. The device is a dual-beam probe-type COD intelligent probe, which effectively reduces turbidity interference through compensation; it comes with a cleaning brush. Avoid the influence of adherents and bubbles on the measurement. The probe adopts RS485 interface and standard Modbus protocol, which is adaptable and easy to use, and it can achieve real-time detection and rapid uploading of COD data in water bodies.



Figure 1.1 Appearance of the COD digital robe

1.2 Measurement principle

This product adopts spectroscopic method to measure COD of water body, including temperature, turbidity and COD three indicators of detection. Temperature is measured by a built-in temperature sensor. The structure of the optical path contains two wavelengths of 254nm and 365nm light source, in which the 254nm wavelength light can be absorbed by the COD in the water, through the establishment of the mathematical model between the absorbance and COD value can be obtained in the water COD value. Since turbidity is also absorbed at 254nm, the probe uses a 365nm light source for simultaneous detection of turbidity in the water, and the COD value is compensated by a compensation algorithm.

In addition, the measured temperature values are temperature compensated for both turbidity and COD values; this greatly improves the accuracy of the measurements.

1.3 Product features

- Self-cleaning function removes light window stains and air bubbles, maintenance-free
- Dual optical path detection, automatic deduction of turbidity interference on COD detection
- Corrosion-resistant housing, waterproof grade IP68, can work underwater for a long time
- RS485 communication interface, standard Modbus protocol for easy integration.

1.4 Main uses and scope of application

Environmental surface water, municipal and industrial wastewater, industrial process water.

2 Technical specification

Working principle	Dual-wavelength ultraviolet absorption		
Meas. Wavelength	254nm, 365nm		
Measurement range	COD: (0~100/200/500) mg/L Other ranges can be customized		Turbidity: 100 NTU (0~100 mg/L) 400 NTU (0~200 mg/L) 600 NTU (0~500 mg/L)
Turb. Comp. range	50 NTU (0~100 mg/L)	100 NTU (0~200 mg/L)	200 NTU (0~500 mg/L)
Repeatability	<1 %		
Accuracy	≤5 %		
Precision	±5 %		
Zero drift (24h)	±2 % F.S. (100 mg/L) ±1 % F.S. (500 mg/L)		
Range drift (24h)	±3 % F.S.		
Detection limit	1 mg/L		
Resolution	0.1 mg/L		
Response time T90	< 30 sec		
Protection class	IP68		
Comm. interface	RS485, standard Modbus protocol, baud rate 9600		
Dimensions	(0-100) mg/L: (φ33*236) mm, cable 10 m (customizable) (0-500) mg/L: (φ33*221) mm, cable 10 m (customizable)		
Materials	316 Stainless-steel/Titanium alloy available		
Weight	1 kg		
Mounting threads	Upper NPT3/4 thread		
Operating voltage	12/24 Vdc		
Power absorption	0.9 W (motor brushes rotating), 0.4 W (motor brushes not working)		
Cleaning method	Mechanical brush		
Calibration period	1~3 months		
Operating temp.	(0~50) °C		

Table 2.1

3 Use, operation

3.1 Installation

The installation of the COD on-line probe (subsequently referred to as the probe) has a greater impact on its reliable measurements, so please note the following requirements:

- When installing, make sure that the light source area is not damaged by external collision, cuts and friction and other factors.
- There should be no obstructions in the installation area to interfere with the optical window of the probe and no external factors to interfere with the normal operation of the brush head.
- The minimum water level should submerge the measuring window area after installation to prevent the water level from being too low and causing abnormal measurement values.
- The sensor measures the water flow velocity of not more than 3.0m/s without abrasive media.
- For submerged installation, the sensor must be mounted on the mounting bracket, and the sensor must not be suspended in water by the sensor cable.
- Considering specific circumstances such as impurity deposition in the water body, it is recommended that the probe be installed horizontally, keeping the measurement window facing vertically downwards.
- In the environment of more debris in the water and high turbidity of the water body, it can be matched with the installation of a protective net cover.

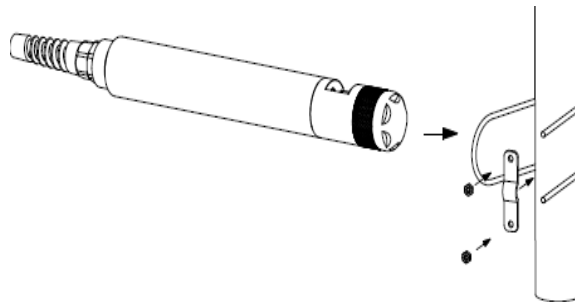


Figure 3.1 Schematic diagram of fixed installation

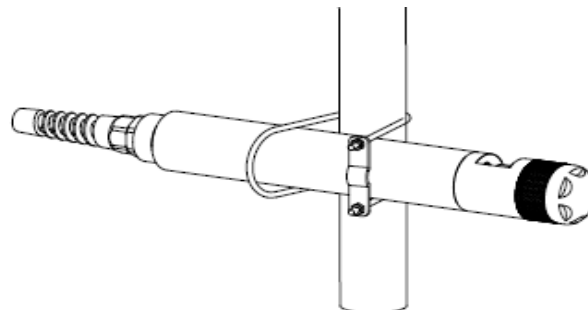


Figure 3.2 Schematic diagram of front mounting

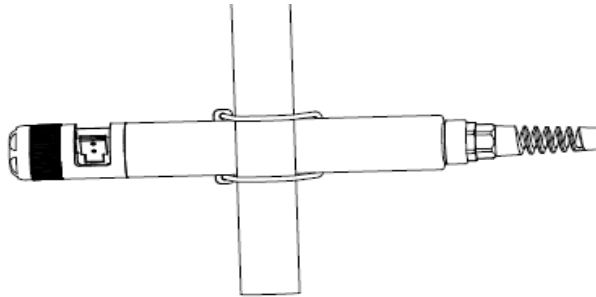


Figure 3.3 Schematic diagram of back mounting

3.2 Wiring instructions

Sensor type	Wire color	Meaning	Connection type
UVCOD200 COD digital probe	red	Power positive (+)	12/24 Vdc power supply
	black	Power negative (-)	12/24 Vdc power supply
	blue	485A	RS-485 communication
	yellow-brown	485B	RS-485 communication

Table 3.1 Wiring description

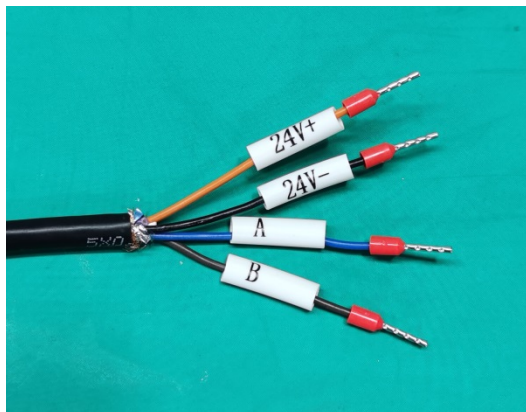


Figure 3.4 Cable wiring

Sensor wiring, strip the end of the cable to reveal 5 wires, please check the table 3.1 were connected to the corresponding port, any short circuit may cause the instrument burned, data loss or program damage! Confirmation of the correct wiring can be connected to the power to use.

3.3 Communication protocol

baud	9600
data bit	8
stop bit	1
parity check	None
slave address	0x32

Table 3.2 Serial Port default Parameter Configuration

Modbus-RTU mode, data transfer size end description:

data type	Big and small end model
char (computing)	the main points
uint16	the main points
uint32	Word-ordered little end, byte-ordered big end within word (CDAB)
Float (encoding format IEEE754)	Word-ordered little end, byte-ordered big end within word (CDAB)

Fault code	Reason for failure
1ul<<0	Motor failure, gear slippage
1ul<<1	Motor failure, blockage
1ul<<2	Zero voltage too low
1ul<<3	COD linear deviation

Table 3.3 Fault table

Note:

<< is a left-shift operator that first converts the alarm code to binary, with each bit representing an alarm state

For example, 44 is converted to 00000000 00000000 00000000 00101100, which corresponds to fault codes 1ul<<2,1ul<<3,1ul<<5

Functionality	Typology	Meas. unit	Range of values	Byte count	Input register address	Note
Device SN number	ASCII	--	--	12	0x7530	
software version number	ASCII	--		32	0x7536	
Probe Type	uint16	--	--	2	0x7546	Fixed at 5 for COD
Probe Status	uint16	--	0-255	2	0x7566	0: idle 1: Measurement 2: under calibration 3: motor rotation
Error code	uint32	--	--	4	0x7567	Normal state is 0, see fault table for details
COD concentration	Float	mg/L	--	4	0x7569	
temp	Float	°C	0-50	4	0x756b	
turbidity	Float	NTU	--	4	0x756d	
254 voltage	Float	V	--	4	0x756f	
356 voltage	Float	V	--	4	0x7571	
Absorbance at 254nm	Float	--	--	4	0x7573	
Absorbance at 365nm	Float	--	--	4	0x7575	
Zero calibration temperature	Float	°C	0-50	4	0x7582	
Zero calibration 365 voltage	Float	V	--	4	0x7584	
Zero calibration 254 voltage	Float	V	--	4	0x7586	
NTU1 calibration 365 voltage	Float	V	--	4	0x7588	
NTU1 calibration 254 voltage	Float	V	--	4	0x758a	
NTU1 calibration 365 absorbance	Float	--	--	4	0x758c	

NTU1 calibration 254 absorbance	Float	--	--	4	0x758e	
NTU1 calibration temperature	Float	°C	0-50	4	0x7590	
NTU2 calibration 365 voltage	Float	V	--	4	0x7592	
NTU2 calibration 254 voltage	Float	V	--	4	0x7594	
NTU2 calibration 365 absorbance	Float	--	--	4	0x7596	
NTU2 calibration 254 absorbance	Float	--	--	4	0x7598	
NTU2 calibration temperature	Float	°C	0-50	4	0x759a	
COD1 calibration 365 voltage	Float	V	--	4	0x759c	
COD1 calibration 254 voltage	Float	V	--	4	0x759e	
COD1 calibration 365 absorbance	Float	--	--	4	0x75a0	
COD1 calibration 254 absorbance	Float	--	--	4	0x75a2	
COD1 calibration temperature	Float	°C	0-50	4	0x75a4	

COD2 calibration 365 voltage	Float	V	--	4	0x75a6	
COD2 calibration 254 voltage	Float	V	--	4	0x75a8	
COD2 calibration 365 absorbance	Float	--	--	4	0x75aa	
COD2 calibration 254 absorbance	Float	--	--	4	0x75ac	
COD2 calibration temperature	Float	°C	0-50	4	0x75ae	
COD3 calibration 365 voltage	Float	V	--	4	0x75b0	
COD3 calibration 254 voltage	Float	V	--	4	0x75b2	
COD3 calibration 365 absorbance	Float	--	--	4	0x75b4	
COD3 calibration 254 absorbance	Float	--	--	4	0x75b6	
COD3 calibration temperature	Float	°C	0-50	4	0x75b8	
TSS value	Float	mg/L	--	4	0x75ba	
BOD value	Float	mg/L	--	4	0x75bc	

Table 3.4 Input register protocols

R/W properties: phrase marked by pause

Functionality	Read and Write Properties	Typology	Meas. unit	Range of values	Occupied byte	Holding Register Address	Note
Device address	Read/Write	uint16	--	0 to 255	2	0x9c40	The default value is 0x32, the device is reset after writing, so it must be written to a single register.
Baud	Read/Write	uint32	--	4800. 9600. 14400. 19200. 38400. 57600. 115200.	4	0x9c41	Default value 9600
Compensation factor a	Read/Write	float	--	--	4	0x9c43	Default 1
Compensation factor b	Read/Write	float	--	--	4	0x9c45	Default 0
Motor operating cycle	Read/Write	float	min	--	4	0x9c47	Default value 20
Sampling interval	Read/Write	uint16	s	--	2	0x9c49	Default value 30
Restore factory settings	Write	uint16	--	0-1	2	0x9c51	Automatic reset after writing
Motor control	Write	uint16	--	0-1	2	0x9c53	
Fault clearance	Write	uint16	--	0-1	2	0x9c54	Performs clearing when writing 1
Range switching	Write	uint16	--	0-1	2	0x9c55	0: high range 1: low range
Restart SW	Write	uint16	--	0-1	2	0x9c56	Automatic reset after writing
TSS_K	Read/Write	float	--	--	4	0x9c5c	
BOD_K	Read/Write	float	--	--	4	0x9c5e	
Temperature calibration time	Read/Write	uint32	s	--	4	0x9c60	UNIX timestamp (calibration starts after writing)

Temperature calibration value	Read/Write	float	--	--	4	0x9c62	Must be written at the same time as the timestamp, calibration starts automatically after successful writing
Zero calibration time	Read/Write	uint32	s	--	4	0x9c64	UNIX timestamp (calibration starts after writing)
Turbidity 1 calibration time	Read/Write	uint32	s	--	4	0x9c66	UNIX timestamp (calibration starts after writing)
Turbidity 1 calibration value	Read/Write	float	--	--	4	0x9c68	Must be written at the same time as the timestamp, calibration starts automatically after successful writing
Turbidity 2 calibration time	Read/Write	uint32	s	--	4	0x9c6a	UNIX timestamp (calibration starts after writing)
Turbidity 2 calibration value	Read/Write	float	--	--	4	0x9c6c	Must be written at the same time as the timestamp, calibration starts automatically after successful writing
COD 1 calibration time	Read/Write	uint32	s	--	4	0x9c6e	UNIX timestamp (calibration starts after writing)
COD 1 calibration value	Read/Write	float	--	0-1000	4	0x9c70	Must be written at the same time as the timestamp, calibration starts automatically after successful writing
COD 2 calibration time	Read/Write	uint32	s	--	4	0x9c72	UNIX timestamp (calibration starts after writing)
COD 2 calibration value	Read/Write	float	--	0-1000	4	0x9c74	Must be written at the same time as the timestamp, calibration starts automatically after successful writing
COD 3 calibration time	Read/Write	uint32	s	--	4	0x9c76	UNIX timestamp (calibration starts after writing)
COD 3 calibration value	Read/Write	float	--	0-1000	4	0x9c78	Must be written at the same time as the timestamp, calibration starts automatically after successful writing

Table.3.5 Holding registers

3.4 MODBUS-RTU protocol example (address default 0x32)

3.4.1 Read the measured value

Send: 32 04 75 69 00 02 be 18

0x32:Device address 0x04: Read input register function code 0x7569:Temperature register address

0x0002: Read two registers

0xbe18: check digit

Received: 32 04 04 75 ce 43 5d 73 bd

0x32:Device address 0x04: Read input register function code 0x04:4 bytes

0x75ce435d:COD 0x435d75ce (221.46)

0x73bd:Check Digit

3.4.2 Read the temperature

Send: 32 04 75 6b 00 02 1f d8

0x32:Device address 0x04: Read input register function code 0x756b:Temperature register address

0x0002: Read two registers

0x1fd8: check digit

Received: 32 04 04 5e a4 41 cb db 4b

0x32:Device address 0x04: Read input register function code 0x04:4 bytes

0x5ea441cb:temperature 0x41cb5ea4 (25.42)

0xdb4b:check digit

3.4.3 Zero calibration

Send: 32 10 9c 64 00 02 04 b7 4f 65 0b 81 c2

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c64:Write register start address

0x02:Write to 2 registers

0x04:write 4 bytes

0xb74f650b:timestamp 0x650bb74f (1695266639) 2023-09-21 11:23:59

0x81c2:Check Digit

Received: 32 10 9c 64 00 02 2b 84

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c64:Write register start address

0x02:Write to 2 registers

0x2b84:check digit

3.4.4 COD2 calibration

Send: 32 10 9c 72 00 04 08 06 1f 64 de 00 00 42 c8 52 a0

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c72:Write register start address

0x04:Write to 4 registers

0x08:Write 8 bytes

0x061f64de:timestamp 0x64de061f (1692272159) 2023-08-17 19:35:59

0x000042c8:Calibration value 0x42c80000(100)

0x52a0:check digit

Received: 32 10 9c 72 00 04 4a 42

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c72:Write register start address

0x04:Write to 4 registers

0x4a42:check digit

3.4.5 Motor operation

Send: 32 06 9c 53 00 01 93 88

0x32:Device address

0x06:Write Single Register Function Code

0x9c53:Write register address

0x0001:Parameter 1

0x9388:Check Digit

Received: 32 06 9c 53 00 01 93 88

0x32:Device address

0x06:Write Single Register Function Code

0x9c53:Write register address

0x0001:Parameter 1

0x9388:Check Digit

4 Use and maintenance

4.1 Calibration

The COD online probe is available in three models, according to the following measurement ranges:

- 0-100 mg/L
- 0-200 mg/L
- 0-500 mg/L.

The probe is connected to the controller or host computer and the calibration is completed according to the following steps

4.1.1 Temperature calibration

Place the clean probe in pure water and at the same time place the thermometer near the probe; after the temperature is stable, write the temperature value in the calibration value register first, and then write the time to start the calibration in the calibration time register first, and wait for 10 seconds to complete the calibration..

4.1.2 Zero calibration

Zero calibration is required first before use. Place the clean probe in pure water, control the brush head to rotate once, and then perform zero calibration. After stabilization, write the time in the calibration register to start calibration and wait for 120 seconds to complete the calibration.

4.1.3 0-100 range turbidity calibration

Using 20 NTU (turbidity 1) and 60 NTU (turbidity 2) turbidity standard solution, when a certain calibration point calibration, using the corresponding standard solution will be lubricated after the probe is placed in the standard solution, the control of the brush head rotated once, and then carry out the calibration of the corresponding point; the calibration process needs to wait for 120 seconds. The same operation is completed for another turbidity calibration point.

4.1.4 0-200 range turbidity calibration

Using 50NTU (turbidity 1) and 200NTU (turbidity 2) turbidity standard solution, when a certain calibration point calibration, using the corresponding standard solution, the probe will be wetted and placed in the standard solution, the control of the brush head rotated once, and then carry out the corresponding point calibration; the calibration process needs to wait for 120 seconds. The same operation is completed for another turbidity calibration point.

4.1.5 0-500 range turbidity calibration

Using 50 NTU (turbidity 1) and 200 NTU (turbidity 2) turbidity standard solution, when a certain calibration point calibration, using the corresponding standard solution, the probe will be wetted and placed in the standard solution, the control of the brush head rotated once and then carry out the corresponding point calibration; the calibration process needs to wait for 120 seconds. The same operation is completed for another turbidity calibration point.

4.1.6 0-100 range COD calibration

The calibration is made on two points; COD calibration solution is obtained by using potassium hydrogen phthalate; 20mg/L (COD1) and 80mg/L (COD2) COD calibration solution is used. When a certain calibration point is calibrated, the probe is lubricated with the corresponding calibration solution, placed in the calibration solution, and the brush head is rotated once, and then the calibration of the corresponding point is carried out; the calibration process needs to wait for 120 seconds. 120 seconds. The same operation is completed for another COD calibration point.

4.1.7 0-200 range COD calibration

The calibration is made on three points; COD standard solution using potassium hydrogen phthalate configuration; using 20mg / L (COD1), 100mg / L (COD2) and 200mg / L (COD3) COD standard solution, when a certain calibration point calibration, using the corresponding standard solution, the probe will be wetted and placed in the standard solution, the control of the brush head to rotate once, and then carry out the corresponding point calibration; calibration process needs to wait for 120 seconds; calibration process needs to wait for 120 seconds. calibration; the calibration process needs to wait for 120 seconds. The same operation is done for another COD calibration point

4.1.8 0-500 range COD calibration

The calibration is made on three points; COD calibration solution is obtained by using potassium hydrogen phthalate; 30mg/L (COD1), 100mg/L (COD2) and 400mg/L (COD3) COD calibration solution is used; when a certain calibration point is calibrated, the probe is lubricated with the corresponding calibration solution, placed in the calibration solution, and then the brush head is rotated once, and the calibration of the corresponding point is carried out; the calibration process takes 120 seconds; the calibration process needs to wait for 120 seconds. calibration; the calibration process needs to wait for 120 seconds. The same operation is done for another COD calibration point.

NOTE: In order to ensure the accuracy of the calibration, it is best to control the temperature difference between the five standard solutions used within 1°C.

4.2 Preparation of calibration solutions

4.2.1 Preparation of 20 fumed silica standard solution

After shaking well the 4000NTU fumed silica mother standard solution, accurately draw 5 mL of this solution into a 1000 mL volumetric flask with a pipette, dilute it to the mark with purified water and shake well. The concentration of this solution is 20 NTU turbidity standard solution.

4.2.2 Preparation of 60 NTU fumed silica standard solution

After shaking well the 4000 NTU fumed silica mother standard solution, use a pipette to accurately draw 15 mL of this solution in a 1000 mL volumetric flask, dilute it to the mark with purified water and shake well. The concentration of this solution is 60 NTU turbidity standard solution.

4.2.3 Preparation of 50 NTU fumed silica standard solution

After shaking well the 4000 NTU fumed silica mother standard solution, accurately pipette 12.5 mL of this solution into a 1000 mL volumetric flask, dilute to the mark with purified water and shake well. The concentration of this solution is 50 NTU turbidity standard solution.

4.2.4 Preparation of 200 NTU fumed silica standard solution

After shaking well the 4000 NTU fumed silica mother standard solution, accurately draw 50 mL of this solution into a 1000 mL volumetric flask with a pipette, dilute it to the mark with purified water and shake well. The concentration of this solution is 200 NTU turbidity standard solution.

4.2.5 Preparation of 20 mg/L COD standard solution

Pipette 20 mL of 1000 mg/L COD calibration solution accurately into a 1000 mL volumetric flask, dilute to the mark with pure water and shake well. This solution COD concentration is 20 mg/L.

4.2.6 Preparation of 30 mg/L COD standard solution

Pipette 30 mL of 1000 mg/L COD calibration solution accurately into a 1000 mL volumetric flask, dilute to the mark with pure water and shake well. This solution COD concentration is 30 mg/L.

4.2.7 Preparation of 100 mg/L COD standard solution

Pipette 10 mL of 10000 mg/L COD standard storage solution into 1000 mL volumetric flask accurately, dilute to the mark with pure water and shake well. The COD concentration of this solution is 100 mg/L.

4.2.8 Preparation of 200 mg/L COD standard solution

Pipette 20 mL of 10000 mg/L COD standard storage solution into 1000 mL volumetric flask accurately, dilute to the mark with pure water and shake well. The COD concentration of this solution is 200 mg/L.

4.2.9 Preparation of 400 mg/L COD standard solution

Pipette 40 mL of 10,000 mg/L COD standard storage solution accurately into a 1000 mL volumetric flask, dilute to the mark with purified water and shake well. The COD concentration of this solution is 400 mg/L.

4.3 Calibration of compensation coefficient of real water samples

Use the standard analytical method and this product to test two groups of representative real water samples with different concentrations of COD under the same conditions, and then get the compensation coefficients of actual water samples, a and b, with a as the slope, by linear fitting with these two groups of values. Then write the values of a and b into the corresponding registers to complete the calibration. a default value is 1, b default value is 0.

As an example, please find here below the measured values of two representative samples, to be measured respectively by the UV COD probe and the laboratory analytical method:

Water samples	UV COD test value	Standard method test value
Water sample 1	26.38	10.56
Water sample 2	11.25	6.67

Table 4.3.1

Figure 4.3.1 shows the establishment of a linear relationship, with the UV COD test values as horizontal coordinates and the standard method test values as vertical coordinates.

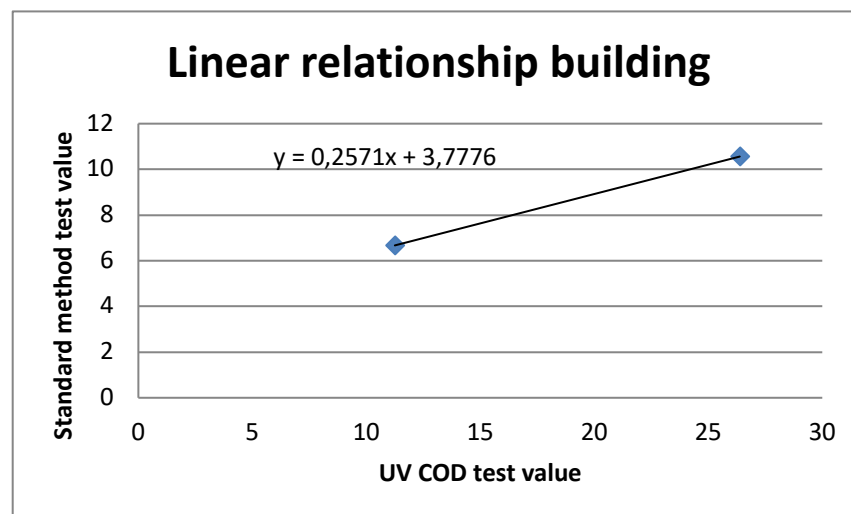


Figure 4.2.1. Linear Relationship Establishment

The compensation factor a = 0.2571 and the compensation factor b = 3.7777.

4.4 Measurement

After calibration, the sensor can measure, store and upload data normally. The motor cleaning cycle can be set according to the water quality, and the water samples can be measured continuously for a long time.

4.5 Maintenance

Maintenance of the probe consists mainly of cleaning and calibration:

Cleaning: In order to prevent impurities and microorganisms from adhering to the surface of the glass window of the light source or clogging of the motor, resulting in inaccurate measurements, it is necessary to clean the product once a week. Remove the sensor product from the water, use a clean non-woven cloth and distilled water to remove the surface and the motor brushes attached to the dirt, be careful not to use excessive force to damage the components.

Calibration: It is usually recommended to perform a calibration once a week after cleaning, to ensure the accuracy of the test.

Wiper replacement:

- power ON: turn on the COD analyzer. When the wiper brush stops in the middle of the optical window, immediately turn off the power;
- loosen Screws: Use an M1.5 hex key to loosen the fixing screws on both sides of the wiper. Use needle-nose pliers to pull out the wiper that needs to be replaced;
- install new wiper: Insert the new wiper. Use the M1.5 hex key to tighten the fixing screws on both sides.

4.6 Troubleshooting

If not according to the user manual correct installation, wiring, improper use or component damage may cause instrument failure, Table 4.5.1 gives the instrument several typical faults and troubleshooting methods. When the instrument has a problem, please firstly compare the table 4.5.1 to see which kind of fault belongs to, and then solve the problems one by one according to the tips; if the problems encountered are more complicated, please contact with our company in time to get support, and don't try to solve the problems by yourself.

Table 4.1 Typical faults and handling

Serial number	Fault phenomenon	Possible causes	Exclusion method
1	Abnormal motor rotation	foreign body obstruction	Firstly, observe whether the motor is hindered by external foreign matter, if there is excluded foreign matter to observe whether to restore, when the use of time is too long motor aging rust can be rust cleaning and then observe whether to restore, or need to return to the factory for repair!
		Motor ageing and rusting	
		motor failure	
2	COD test value is 0 or very large	excessive turbidity of water samples	First judge whether the turbidity value exceeds the range; then observe whether the two light sources are normal light, whether the outside world is blocked; if still can not be solved need to return to the factory for maintenance
		light source abnormality	
		Signal processing abnormality	

5 Additional information

5.1 Safekeeping

When the instrument you purchased is not to be installed urgently or needs to be put aside for a longer period of time, please store it properly and do not put it in the following places:

- Ambient temperature below 5°C or above 40°C;
- Direct sunlight;
- Next to the heater;
- The environment is vibrating and dusty;
- When the instrument you have purchased has been installed but will not be used for a longer period of time, wipe the probe dry and store it in a cool, dry place.

5.2 Precautions to use

★ Avoid using the motor with too short a rotation cycle.

★ When using it, you need to avoid water cut-offs that cause impurities to remain on the glass window and scratching the glass exterior when the motor rotates.

5.3 Unpacking and inspection

When you receive the purchased instrument to open the box before the inspection, please carefully check the packaging for any damage or signs of falling, if there is any suspicion and cause damage to the contents of the goods, quickly notify the company and the carrier! Please be careful when opening the box, do not damage the instrument or other parts. After opening the box, please check each piece, if you find the wrong shipment, missing or damaged, please notify the company as soon as possible.

5.4 After sale service

When during the installation, commissioning and use of the process would be encountering problems, please consult our company in a timely manner, especially found fault, please as far as possible will be the instrument number, model number, date of purchase, failure phenomenon described clearly so that we can quickly and accurately to provide you with support.